

HEAIN Core Protocol: A Normative Specification for Collective Intelligence in Fractal Edge AI Governance

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Abstract

Hybrid Edge AI Node establishes a fractal governance architecture for distributed AI infrastructure, but as a theoretical contribution it leaves the exact wire-level and operational behavior of a conformant implementation underspecified. This paper introduces HEAIN Core Protocol (HEAIN-CORE-1.0), a normative, RFC-2119-style specification that operationalizes that architecture for implementers. Three contributions distinguish this specification from the parent architecture paper. First, a three-layer positioning clarifies where "intelligence" resides in the system: Hybrid Edge AI Node is the architecture, HEAIN Core Protocol is an infrastructure-level collective and adaptive intelligence that animates it through swarm learning and self-healing, and any organization's production models (e.g., CineNexus Pro's image, video, and audio models) are application-level AI executed within one procedure of the protocol. Second, the protocol extends the previously binary Master/Worker role model into a symmetric four-tier hierarchy (Worker, Zone, Secondary, Global Primary) governed by a One-Tier-At-A-Time promotion rule, which provably prevents a node from assuming authority over a scope larger than its immediate context even under cascading, multi-tier failure. Third, a Configuration API specification closes a governance gap left open in the architecture paper: administrative access is split into operational parameters that an authenticated administrator may change directly, and policy parameters that must be routed through the protocol's own Policy Boundary Check procedure before taking effect, with every change — regardless of category — recorded in an append-only audit log. We present the protocol's envelope format, node bootstrap and dual-loop execution model, and the governance rules for its seven core procedures, and situate it as a companion, implementation-facing specification to the peer architecture paper.

Keywords: Protocol Specification, Distributed Systems, Fractal Hierarchy, Configuration Governance, Collective Intelligence, Edge Computing, RFC 2119

1. Introduction

A preceding paper, Hybrid Edge AI Node: A Fractal Governance Architecture for Sovereignty-Aware Distributed AI Processing [1][2], established the theoretical case for a distributed AI infrastructure whose control-plane hierarchy grows and contracts with an organization's size, whose autonomous decisions are bounded by a human-governed policy layer with an explicit emergency exception, and whose nodes can exchange learned knowledge across the network without violating jurisdictional data-residency requirements. That paper answers "why"; it does not answer, at the level required for interoperable implementation, "how exactly." Two independent teams implementing the architecture from the theoretical paper alone would very likely produce systems that cannot exchange messages, because no common wire format, node addressing scheme, or administrative-access model was defined.

This paper addresses that gap by introducing HEAIN Core Protocol (identifier HEAIN-CORE-1.0), a companion specification written in the normative style of Internet Engineering Task Force documents, using the requirement-level keywords **MUST**, **SHOULD**, and **MAY** as defined in RFC 2119 [3]. Beyond simply formalizing the architecture paper's seven algorithms, this specification resolves three questions the architecture paper left open, each of which required a nontrivial design decision: where does "intelligence" actually reside in a system whose individual procedures are deterministic; how does role promotion generalize from a two-role (Master/Worker) model to the fully recursive four-tier hierarchy the architecture paper's fractal principle implies; and how is the protocol's own configuration governed, given that an administrative interface which can freely rewrite policy thresholds would silently defeat the human-oversight mechanism the architecture paper was designed to guarantee.

2. Related Work

This paper builds directly on the architecture and literature review of the parent paper [1], which situates the six governing mechanisms of Hybrid Edge AI Node against KubeEdge and Greengrass-style edge orchestration, hierarchical federated learning, swarm computing and digital immune systems, Raft/Paxos/Bully leader election, human-in-the-loop AI governance frameworks, and sovereign edge computing. Rather than repeat that review, this paper's related work is scoped to protocol specification practice itself. The Internet Engineering Task Force's RFC 2119 [3] established the convention of capitalized requirement-level keywords now used across virtually all Internet protocol specifications; this paper adopts that convention directly so that HEAIN Core Protocol can be read and implemented using the same interpretive discipline as, for example, TCP or HTTP specifications. Configuration-governance separation of the kind introduced in Section 7 has informal precedent in role-based access control (RBAC) systems generally, but RBAC alone does not address the specific problem this paper solves: distinguishing configuration fields that are safe for direct administrative modification from configuration fields that are themselves inputs to the system's own governance procedure, and routing the latter through that procedure rather than around it.

3. Protocol Architecture Overview

HEAIN Core Protocol operates across the same four layers as the parent architecture — Management & Orchestration, Edge Execution, Swarm Intelligence, and Network & Security — and implements seven core procedures (HEAIN-P1 through P7) corresponding to the parent paper's seven algorithms. Every message exchanged between nodes **MUST** be transported over mTLS at the Network & Security layer without exception. Nodes are addressed using a hierarchical URI scheme that mirrors the fractal governance structure directly: `heain://<organization>/<country>/<zone>/<node-id>`, with higher tiers omitting the segments below their scope (e.g., `heain://cinexus/primary` for a Global Primary or Primary/Secondary node). A common Envelope data structure carries every unit of work or inter-node message; its required fields (`ticket_id`, `encrypted_payload`, `staging_flag`, `retry_count`, `origin_zone`, `timestamp`) and optional fields (`category`, `generation`) are reused unmodified across all seven procedures, giving implementers a single serialization format to support rather than seven.

4. Three-Layer AI Positioning

A recurring source of ambiguity in early drafts of this work was the question of where the system's "intelligence" is located, since every individual procedure (P1–P7) is specified as deterministic, rule-based logic with no embedded machine-learning inference. We resolve this by distinguishing three layers explicitly.

Layer	Component	Role
1 — Architecture	Hybrid Edge AI Node	The theoretical, published architecture [1][2]
2 — Infrastructure AI	HEAIN Core Protocol	Collective, adaptive intelligence that animates the architecture
3 — Application AI	Deployer's own models (e.g., CineNexus Pro)	Image/audio/video models executed inside procedure P3

Table 1. Three-layer positioning of intelligence within the HEAIN ecosystem

Hybrid Edge AI Node is the architecture; HEAIN Core Protocol is the collective intelligence that brings it to life; each deployer's Application AI runs on top of that living infrastructure.

The protocol's claim to "intelligence" at layer 2 rests specifically on two mechanisms operating at network scale rather than within any single node: knowledge propagation (P7, Sovereignty-Aware Swarm Broadcast), through which a discovery made by one node improves the behavior of every node in the network, and self-directed adaptation (P6, Role Transformation), through which the network reorganizes its own control structure in response to failure without waiting for human intervention. Both mechanisms remain, however, strictly bounded by P5 (Policy Boundary Check), which is precisely what allows the network's adaptive behavior to be described as governed collective intelligence rather than uncontrolled emergent behavior — the distinction the AI-governance literature reviewed in the parent paper identifies as the central design problem for autonomous systems generally.

5. Node Lifecycle: Bootstrap and Dual-Loop Execution

Every conformant node, regardless of tier, MUST execute an identical bootstrap sequence: register its `heain://` address, establish its mTLS channel, retrieve the network's current generation number (Section 6), and start two background listeners — a heartbeat emitter directed at its parent, and a Configuration API listener (Section 7). Following bootstrap, execution diverges into exactly one of two loops selected by the node's current tier: a Worker Loop for tier WORKER, or a parameterized Master Loop for tiers ZONE, SECONDARY, and GLOBAL_PRIMARY. Only a ZONE-tier Master Loop instance directly dispatches jobs to Workers (procedure P2); Master Loop instances at SECONDARY or GLOBAL_PRIMARY tier that receive a client request MUST route it down to the appropriate Zone before dispatch can occur. This single-Main-Loop, tier-parameterized formulation makes explicit a property the architecture paper states but does not operationalize: that Master and Worker are not distinct programs but two execution modes of the same node, selected by a single configuration value (tier) that Section 6's promotion procedure is permitted to change at runtime.

6. One-Tier-At-A-Time Promotion

The architecture paper's Role Transformation mechanism permits a Worker Node to assume Master responsibilities during failure, guarded by a generation-number fencing scheme that prevents split-brain conditions. That mechanism was originally specified only for the two-tier Master/Worker case. Generalizing it to the four-tier hierarchy introduced a hazard the architecture paper did not anticipate: without an explicit constraint, a cascading failure that removed a Global Primary and every Secondary Master simultaneously could, under a naive generalization, permit a Worker Node to be promoted directly to Global Primary in one step — assuming authority over an entire multinational organization from a node whose stability score reflects only its performance as a single job-processing unit.

HEAIN Core Protocol closes this hazard with a single rule: promotion candidates for procedure P6 MUST be drawn exclusively from nodes at the tier immediately adjacent to the failed node — siblings at the same tier, or direct children one tier below — and a promoted node MUST advance by exactly one tier per invocation of P6, never more.

```
WORKER    -> ZONE          (candidates: sibling Workers in the same zone)
ZONE      -> SECONDARY     (candidates: sibling Zone Masters in the same country)
SECONDARY -> GLOBAL_PRIMARY (candidates: sibling Secondary Masters org-wide)
```

Even under the worst-case scenario described above, a Worker Node MUST NOT be promoted directly to Global Primary; it can only be promoted to Zone Master, after which a subsequent, independent invocation of P6 at the next tier up would be required before authority could propagate further. This constraint trades a small amount of failover latency in catastrophic, multi-tier outage scenarios for a guarantee that no node ever assumes authority over an operational scope larger than one step beyond what its own monitoring and stability history can meaningfully reflect.

7. Configuration API and Governance-Aware Administration

The architecture paper's Policy Boundary Check procedure (P5) guarantees that autonomous system behavior remains within a board-defined boundary. That guarantee is only as strong as the mechanism used to set the boundary in the first place; an administrative interface capable of directly rewriting policy thresholds would allow the same governance boundary to be silently widened through an operational back door, entirely outside the audit and approval discipline P5 otherwise enforces. HEAIN Core Protocol closes this gap by specifying a two-tier permission model for its Configuration API.

Tier	Rule	Example fields
1 — Access gate	Every endpoint MUST require authenticated administrator identity	All endpoints
2a — Operational	Administrator may modify directly, statically or dynamically	heartbeat_interval, MAX_RETRY, lease_duration, assembly_timeout
2b — Policy	Administrator cannot modify directly; the request MUST be submitted as a proposed_action to P5 before taking effect	policy thresholds, allowlist categories

Table 2. Two-tier Configuration API permission model

Every configuration change — operational or policy, static or dynamic, approved automatically or after human review — MUST be written to the same append-only audit log that P5 itself writes to, recording the administrator identity, the field changed, its prior and new values, and a timestamp. This closes

what would otherwise be an asymmetry in the architecture's transparency guarantee: without this requirement, autonomous decisions made by the protocol are fully auditable while human administrative decisions are not, undermining the accountability claim the parent paper makes for the system as a whole. With it, every change to system behavior, whether initiated by the protocol's own governance logic or by a human administrator, is recorded in the same ledger and subject to the same retrospective scrutiny.

8. Core Procedures

HEAIN-P1 through P7 correspond directly to the seven algorithms specified in the architecture paper [1][2], restated here as normative MUST/SHOULD/MAY requirements rather than descriptive prose; the full requirement-level text for each procedure is maintained in the companion protocol specification document referenced in the Data Availability statement below. Conformance is defined in two levels: an implementation supporting P1–P5 satisfies Core Conformance, sufficient for a single-zone deployment with no role transformation or cross-zone knowledge sharing; an implementation supporting all seven procedures satisfies Full Conformance, required for any deployment with more than one Worker Node or more than one Zone.

9. Discussion

This specification is deliberately scoped as an implementation-facing companion to the architecture paper rather than a replacement for it: Sections 4, 6, and 7 above are original contributions made necessary by the act of specifying the architecture precisely enough to implement, and were not present in the original theoretical treatment. This pattern — a theoretical architecture paper followed by a separate, narrower protocol specification — mirrors the historical relationship between distributed-systems research papers (e.g., the original Raft paper) and their subsequent formal protocol write-ups, and is intended to let each document be evaluated on its own terms: the architecture paper for its novelty and positioning relative to prior research, this specification for its precision and implementability.

The specification inherits the empirical limitations disclosed in the architecture paper without modification: no procedure described here has been benchmarked under production load, and the reference implementation (`cinexus_master.py` / `cinexus_worker.py`) currently covers only a partial subset of Procedures 1–3. The Configuration API and One-Tier-At-A-Time promotion rule introduced in this paper are, at the time of writing, unimplemented design requirements rather than validated engineering results, and should be read accordingly.

The empirical benchmarking this specification still requires is methodologically consistent with the author's earlier peer-reviewed work comparing block-access (iSCSI) and file-access (CIFS/Samba) I/O throughput under controlled network conditions using the Iozone, PostMark, and SQLio benchmark tools [5]. That study's central finding — that protocol performance is highly dependent on workload shape and access pattern rather than uniformly favoring one design — motivates evaluating each of HEAIN Core Protocol's seven procedures under multiple distinct workload profiles rather than a single synthetic load, once a full reference implementation is available.

10. Conclusion

HEAIN Core Protocol operationalizes the Hybrid Edge AI Node architecture into a normative, implementable specification, resolving three questions left open by the theoretical treatment: a three-layer account of where intelligence resides in a system built from deterministic procedures, a symmetric four-tier promotion rule that generalizes role transformation safely beyond the original two-role model, and a governance-aware configuration model that prevents administrative access from becoming an unaudited bypass of the architecture's own policy-enforcement mechanism. Future work should prioritize a reference implementation covering all seven procedures, empirical benchmarking of the promotion and swarm-broadcast procedures under simulated multi-tier failure, and, in time, transition of this specification's governance model from author-controlled to a benevolent-maintainer process open to external review, as discussed but deliberately deferred during the design of this version.

Data Availability: The full normative text of HEAIN-CORE-1.0, including complete pseudocode for all seven procedures, is maintained as a companion technical specification document alongside this paper.

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